

CSCI235 – Database Systems
2026 S2
Implementation Task 1
Due on 26 April 2026 by 9:00 pm SGT

Scope

The Implementation of Task 1 is related to the contents of topic on indexing.

This Implementation is due by Sunday, 26 April 2026, 9:00 pm Singapore time.
This task is worth 3% of the total assessment for the subject.

Only electronic submission through Moodle at:

<https://moodle.uowplatform.edu.au/>

is accepted. All email submission will be deleted and mark 0 ("zero") will be awarded.

The submission procedure is explained at the end of this specification.

Specification

Task 1 (1.5 marks)

1. Consider a relational schema $R(A, B, C, D, E)$ and the following set of functional dependencies valid in the schema:

(i) $\{A \rightarrow B, C \rightarrow D, E \rightarrow A\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

Sample solution includes but not limited to the following:

If $E \rightarrow A$ and $A \rightarrow B$ then $E \rightarrow B$

If $E \rightarrow A$ and $E \rightarrow B$ then $E \rightarrow AB$

If $E \rightarrow AB$ then $ED \rightarrow ABD$

If $C \rightarrow D$ then $EC \rightarrow ED$

If $EC \rightarrow ED$ and $ED \rightarrow ABD$ then $EC \rightarrow ABD$

Minimal key (EC)

(ii) $\{A \rightarrow B, B \rightarrow CD, C \rightarrow E, E \rightarrow AC\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

Sample solution includes but not limited to the following:

If $A \rightarrow B$ and $B \rightarrow CD$ then $A \rightarrow CD$

If $A \rightarrow B$ and $A \rightarrow CD$ then $A \rightarrow BCD$

If $C \rightarrow E$ then $BCD \rightarrow E$

If $A \rightarrow BCD$ and $BCD \rightarrow E$ then $A \rightarrow E$

If $A \rightarrow E$ and $A \rightarrow BCD$ then $A \rightarrow BCDE$

Minimal key (A)

If $C \rightarrow E$ then $CD \rightarrow DE$

If $B \rightarrow CD$ and $CD \rightarrow DE$ then $B \rightarrow DE$

If $C \rightarrow E$ and $B \rightarrow DE$ then $B \rightarrow CDE$

If $E \rightarrow AC$ then $CDE \rightarrow ACD$

If $B \rightarrow CDE$ and $CDE \rightarrow ACD$ then $B \rightarrow ACD$

If $B \rightarrow CDE$ and $B \rightarrow ACD$ then $B \rightarrow ACDE$

Minimal key (B)

If $C \rightarrow E$ and $E \rightarrow AC$ then $C \rightarrow AC$ and $C \rightarrow A$

If $C \rightarrow A$ and $A \rightarrow B$ then $C \rightarrow B$

If $B \rightarrow CD$ then $B \rightarrow D$

If $C \rightarrow B$ and $B \rightarrow D$ then $C \rightarrow D$

If $C \rightarrow E$ and $C \rightarrow A$ and $C \rightarrow B$ and $C \rightarrow D$ then $C \rightarrow ABDE$

Minimal key (C)

If $E \rightarrow AC$ then $E \rightarrow A$

If $E \rightarrow A$ and $A \rightarrow B$ then $E \rightarrow B$

If $E \rightarrow B$ and $B \rightarrow CD$ then $E \rightarrow CD$

If $E \rightarrow AC$ and $E \rightarrow B$ and $E \rightarrow CD$ then $E \rightarrow ABCD$

Minimal key (E)

(iii) $\{A \rightarrow B, DB \rightarrow C\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

Sample solution includes but not limited to the following:

If $A \rightarrow B$ then $AD \rightarrow DB$

If $AD \rightarrow DB$ and $DB \rightarrow C$ then $AD \rightarrow C$

If $AD \rightarrow C$ and $AD \rightarrow BD$ then $AD \rightarrow BCD$

If $AD \rightarrow BCD$ then $ADE \rightarrow BCD$

Minimal key (ADE)

Deliverable

A file Task1Solution.pdf with the solutions of the problems included in Task 1.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

Task 2 (2.0 marks)

Read the following specification of sample database domain.

A hospital would like to create a medical database with the health records of its patients. Information recorded in the database supposed to include a medical insurance number that identifies each patient, first name, last name, date of birth, a list of admission dates to the hospital, list of illnesses cured during each admission and a list of children. Assume, that the first name, last name and date of birth uniquely identify each patient. Assume, that more than one illness can be cured during each admission. Also assume, that child is described by the first name and relation type to its parent, i.e. either son or daughter and all children of a patient have different names.

A database designer created the following relational schema.

MRECORD(insurance,fname,lname,dob,adate,illness,child,relation)

Determine what is the highest normal form the relational schema conforms to ? Justify your choice.

Your next task is to decompose the relational schema into the smallest number of relational schemas, and each of the relational schema is normalized.

Deliverable

A file solution1.pdf with a list of the relational schemas obtained from the decomposition. Each relational schema must have attached a list of nontrivial

functional and/or multivalued dependencies valid in the schema and a justification why the schema is normalized.

Sample solution includes but is not limited to the following.

Obviously, the relational schema above is not in 2NF and it is easy to show it. A functional dependence $\text{insurance} \rightarrow \text{fname}$ is valid in a schema and insurance is not a minimal key because the same insurance number can be combined with many different admission dates (adate). Hence a nonprime attribute fname depends on a subset of a minimal key.

A relational schema MRECORD can be decomposed into the following schemas.

PATIENT(insurance,fname,lname)

ADMISSION(insurance,adate,illness)

CHILDREN(insurance,child,relation)

The following functional dependencies are valid in a relational schema PATIENT.

$\text{insurance} \rightarrow \text{fname}, \text{lname}$

Minimal key = (insurance)

The relational schema PATIENT is in BCNF because the left-hand side of the only functional dependency is a superkey.

The relational schema PATIENT is in 4NF because no nontrivial multivalued dependencies are valid in the schema.

No functional dependencies are valid in a relational schema ADMISSION.

Minimal key = (insurance,date,illness)

The relational schema **ADMISSION** is in BCNF because no functional dependencies are valid in the schema.

The relational schema **ADMISSION** is in 4NF because no nontrivial multivalued dependencies are valid in the schema.

The following functional dependencies are valid in a relational schema **CHILDREN**.
insurance,child $\rightarrow$ relation

Minimal key = (insurance,child)

The relational schema **PCILDREN** is in BCNF because the left-hand side of the only functional dependency is a superkey.

The relational schema **CHILDREN** is in 4NF because no nontrivial multivalued dependencies are valid in the schema.

Submissions

This assignment is due by 9:00 pm (21:00 hours) Sunday, 26 April 2026, Singapore time.

Submit the files **IT1SolutionOutput.txt** through Moodle in the following way:

- 1) Access Moodle at **<http://moodle.uowplatform.edu.au/>**
- 2) To login use a Login link located in the right upper corner the Web page or in the middle of the bottom of the Web page
- 3) When successfully logged in, select a site CSCI235 (SP226) Database Systems
- 4) Scroll down to the section "Implementation Task 1 Submission link" and click at the link.
- 5) Click at a button Add Submission
- 6) Move the file that you want to submit into an area of submission. If you have more than one files to submit, you can drag and drop the remaining files here to add them. You can also use a link *Add...*

- 7) Click at the button "Save changes",
- 8) Click at the check box to confirm authorship of your submission and click at the button "Continue".
- 9) When you are satisfied, remember to click at a button Submit assignment.

A policy regarding late submissions is included in the subject outline. Only one submission per student is accepted.

Implementation Task 1 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.